

“Technologies in Context”

Program Guide for IEEE GHTC 2025

Colorado School of Mines, October 22-25, 2025

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Patrons and Sponsors

Platinum Patrons

The IEEE Humanitarian Technologies Board (HTB) inspires and empowers IEEE volunteers around the world carrying out and supporting impactful humanitarian technology activities at the local level. The IEEE HTB mission is to support impactful and ethically informed volunteer-led initiatives, programs and projects, and mutually beneficial partnerships, as well as to inform policy formulation that harness technology and innovation to address societal challenges (including disaster recovery) in a responsive, effective, and sustainable way.

IEEE Humanitarian Technologies

Humanitarian ENGINEERING
SOCIAL JUSTICE COMMUNITY SUSTAINABILITY
COLORADO SCHOOL OF MINES

The Humanitarian Engineering Program educates engineers and scientists to work as partners with communities seeking to enhance their social, environmental, and economic sustainability. Students in the program learn to take a sociotechnical approach to make a difference in the world through their professional careers, whether in community development or corporate settings. Graduate students and faculty also inform humanitarian engineering education by conducting leading research at the intersection of engineering and social science.

Gold Patron

The Masters of Development Engineering is an interdisciplinary graduate program preparing students to develop scalable and innovative solutions to complex challenges. Development Engineering (DevEng) at Berkeley merges engineering with social science, business, and economics to develop technological interventions that address the needs of low-income communities around the world. Development Engineering differs from traditional engineering by:

UC Berkeley Development Engineering

- Combining social sciences with technical training
- Focusing on social impact: designing for sustainability and practicality first
- Emphasizing human-centered design
- Fostering a deep understanding of the context and culture in which engineers are working

Bronze Patron

eMi
designing a world of hope

EMI is a Christian nonprofit organization founded on the idea that architects, engineers, and design & construction professionals have an important part to play in world missions: We design and build projects that bring hope to communities around the world.

Financial Sponsors

Technical Sponsors

Our vision is to serve the U.S. IEEE member by being the technical professional's best resource for achieving lifelong career vitality and by providing an effective voice on policies that promote U.S. prosperity.

Patron and Sponsor Representatives

Thank you to the representatives of our sponsors and patrons for supporting this conference and ensuring its continued success!

IEEE Humanitarian Technologies Board (HTB):
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Welcome from the Organizers

Welcome from the Conference Chair:



Marie Stettler Kleine
Colorado School of Mines

Welcome to the 15th annual iteration of IEEE GHTC! I have learned so much through my time serving as the conference's general chair. Originally trained as a mechanical engineer, now someone who works to contextualize and complicate motives and values within humanitarian engineering as a science and technology studies scholar, it has been fun and generative to think through organizing this conference as an act of scholarship and service.

But I didn't do it alone. I'll forever be indebted to GHTC 2025's two technical program co-chairs, Grace Burleson and Kirsten Dodson, whom I didn't give ample warning before they committed to volunteering for the gig. In addition, Emeritus Professor Kevin Moore kept us on track as treasurer and operations subcommittee chair. Many thanks to Ed Perkins for his EDAS and publishing support; and it absolutely would not have happened without the onsite close-out crew in Kim Walker, Samantha Temple, Samantha Cesarini, Chelsea Salinas, and Becky Buschke. Special thanks to our sponsors and patrons (detailed on the previous pages), and to our keynote speaker Ann-Perry Witmer whose work speaks directly to the conference theme. It has been generative for me to think with this conference as a scholar interested in "doing good" with and in engineering. I hope participants and attendees see the care that I aspired to bring to the role and this community.

Welcome from the Technical Program Co-Chairs:



Grace Burleson
University of Colorado Boulder



Kirsten Dodson
Lipscomb University

Welcome to GHTC 2025! This year's program features nearly 40 posters and over 100 technical presentations spanning the full spectrum of humanitarian engineering: from innovative technologies to studies of design processes, community partnerships, and engineering education. Together, these contributions reflect the depth and diversity of our field.

This year's program features 18 technical sessions that illustrate the breadth of humanitarian engineering. Innovations in healthcare and disaster prevention/response represent our largest areas. Sessions also explore engineering education, community-centered innovation, and applications in energy, health, education, and agriculture. Several sessions also examine design practice, emphasizing participatory methods, contextual awareness, and meaningful engagement with the communities we partner with and serve.

We extend our heartfelt thanks to all authors, reviewers, and track chairs for their dedication and contribution to this year's program. We hope that GHTC 2025 sparks new insights, collaborations, and inspiration as we continue to explore how "technologies in context" can serve humanity responsibly and equitably.

Organizing Committee

Position	Name
Chair	Marie Stettler Kleine, *Colorado School of Mines*
Technical Program Co-Chair	Grace Burleson, *University of Colorado Boulder*
Technical Program Co-Chair	Kirsten Dodson, *Lipscomb University*
Finance Chair, Treasurer, & Sponsors	Kevin Moore, *Colorado School of Mines (Emeritus)*
Planning & Website	Ed Perkins
Sponsors Committee	Joseph Wei, *IEEE*
Registration Chair, Publicity	Scott Tamashiro, *Raytheon*
Local Arrangement Chair	Caitlin Grabowski, *Colorado School of Mines*
Student Outreach	Samantha Temple, *Colorado School of Mines*

Session Chairs

Thank you to our session chairs for organizing presenters, moderating questions, and making this a collaborative and engaging conference!

Alina Handorean, *Colorado School of Mines*
Anastasia Schauer, *University of Texas at Austin*
Ann-Perry Witmer, *University of Illinois Urbana-Champaign*
Cindy Athanasiou, *Colorado School of Mines*
Dean Nieusma, *Colorado School of Mines*
Elizabeth Reddy, *Colorado School of Mines*
Henry Louie, *Seattle University*
John Gershenson, *Pennsylvania State University*
Juan Lucena, *Colorado School of Mines*
Julia Kramer, *University of Michigan*
Junko Munakata Marr, *Colorado School of Mines*
Khanjan Mehta, *Lehigh University*
Laura MacDonald, *University of Colorado Boulder*
Min Pack, *Baylor University*
Patrick Sours, *Ohio State University*

Student Competition Poster Judges

Thank you to our poster judges for providing critical and thoughtful judging for our student poster competition!

Anastasia Schauer, *University of Texas at Austin*
Ann-Perry Witmer, *University of Illinois Urbana-Champaign*
Chelsea Salinas, *Colorado School of Mines*
Henry Louie, *Seattle University*
James Harper, *University of Colorado Boulder*
John Gershenson, *Pennsylvania State University*
Khanjan Mehta, *Lehigh University*
Laura MacDonald, *University of Colorado Boulder*
Mark Henderson, *University of Colorado Boulder*
Patrick Sours, *Ohio State University*

Technical Paper Reviewers

Thank you to our reviewers for their willingness to provide thoughtful and critical feedback to our authors and presenters!

Aditi Rao
Alberto Martinetti
Alexandria Rodgers
Allan Baez Morales
Anastasia Schauer
Anish Roy
Annalicia Pickering
Anshul Gandhi
Asmah Bohari
Bandar Alzahrani
Bhaskar Bhowmick
Brita Nielsen
Carina Koutsambasis
Chelsea Okwechime
Cristina Mayr
Damien Frame
Daniel Sitompul
Diana Rodriguez Coca
Dongbin Lee
Drishti Marwaha
Eileen Kandie
Elizabeth Ehrhardt
Emily Esteves
Emma Balevic
Eric Lutters
Eric Obeysekare
Franz Busse
Grace Burleson
Han Kyul Kim
Hernan Garcia Arias
Jesus Vargas Bonilla
Jimmy Cordova
Jochebed Basil
John Borland
John Gershenson
Jordan Townsend
Joseph Decuir
Joseph Menicucci
Juan Lucena
Julia Kramer
Julian Zobel
Justin Henriques
Karen Fortuna
Kennedy Orwa
Kevin Moore
Khanjan Mehta
Kirsten Dodson
Krishna Bathula
L.M. Frey
Laura MacDonald
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Maria Mendoza
Marie Ndiaye
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Megan Thomas
Michael DeBellis
Mikayla Friday
Mike Teodorescu
Min Pack
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Papa Alioune Cissé
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Pratyusha Kiran
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Qichao Wang
Rabia Khan
Rahul Autade
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Ruturaj Patil
Rytis Paškauskas
Sajithkumar J
Salim Ghettas
Samantha Sandhaus
Sanket Junagade
Sanyka Banerjee
Stefan Haugen
Stephanie Claussen
Swagatam Bose Choudhury
Taylor Ericson
Thiago Martins de Moraes
Tito Alberto Nuncira Gacharna
Todd Osborn
Tony McAleavy
Victoria Jaqua
Victoria Ustenko
Wilfredo Lugo Beauchamp
Xuanhong Cheng
Yuezi Wu
Yujing Wang
Zahra Rizvi

Schedule at a Glance

WEDNESDAY, OCTOBER 22nd			
2:30 PM	**Registration Opens** *Green Center Lobby*		
3:15 - 4:45 PM	**Workshop 1:** **Centering Context and Communities in IEEE Tech4Good,** Juan Lucena & Diana Duarte *Friedhoff 1*	**Workshop 2:** **Becoming an Intentional Engineer,** Jeff Perry *Friedhoff 2*	
5:00 - 7:00 PM	**Welcome Gathering & Poster Session** *Green Center Lobby*		
THURSDAY, OCTOBER 23rd			
7:30 - 8:00 AM	**Breakfast** *Friedhoff 1*		
8:00 - 8:45 AM	**Conference Welcome & Opening Remarks** *Friedhoff 1*		
8:45 - 9:00 AM	**Networking Break 1**		
9:00 - 10:15 AM	**Session 1A:** *Friedhoff 1* **People-Focused Practice**	**Session 1B:** *Friedhoff 2* **Agriculture & Water**	**Session 1C:** *Friedhoff 3* **Disease Prevention & Care**
10:15 - 10:30 AM	**Networking Break 2**		
10:30 AM - 12 PM	**Session 2A:** *Friedhoff 1* **Disaster Prevention & Response 1**	**Session 2B:** *Friedhoff 2* **Community-Centered Education**	**Session 2C:** *Friedhoff 3* **Context Driven Approaches**
12:00 - 12:30 PM	**Lunch Buffet** *Friedhoff 1*		
12:30 - 1:15 PM	**Panel Session: When Hardware Meets Software - How Humanitarian NGOs Balance Social Dimensions with Technical Interventions** *Friedhoff 1* Sponsored by Mines Humanitarian Engineering Program– Platinum Patron Facilitated by Sam Temple, Colorado School of Mines		
1:15 - 1:45 PM	**Networking Break 3**		
1:45 - 3:15 PM	**Session 3A:** *Friedhoff 1* **Participatory Design**	**Session 3B:** *Friedhoff 2* **Healthcare 1**	**Session 3C:** *Friedhoff 3* **Economic Empowerment**
3:15 - 3:30 PM	**Networking Break 4**		
3:30 - 5:00 PM	**Special Session:** **Current US Policy Trends Shaping Technology** (Erik Heilman, IEEE USA) *60 minutes Friedhoff 1*	**Workshop 3:** **Grid Down Scenario - Table Top Exercise** (David Vaughn, Clemson) *90 minutes Friedhoff 2*	**Special Session:** **Colorado School of Mines Labriola Innovation Tour** *60 minutes* *Meet in Green Center Lobby*
5:00 - 5:30 PM	**Networking Break 5**		
5:30 - 7:30 PM	**Dinner Banquet** *Friedhoff 1* **Keynote by Dr. Ann-Perry Witmer, University of Illinois** **"A Self-ish Approach to Technology in Context"**		

FRIDAY, OCTOBER 24th			
7:30 - 8:15 AM	**Breakfast**		
8:15 - 9:00 AM	**Featured Session: Educational Resources to Deploy Technology for Social Impact and Their Gaps - An Interactive Workshop** *Friedhoff 1* Sponsored by IEEE HTB – Platinum Patron Led by Stephanie Gillespie, IEEE Educational Activities Board		
9:00 - 9:15 AM	**Networking Break 6**		
9:15 - 10:15 AM	**Session 4A:** *Friedhoff 1* **Data-Driven Decisions**	**Session 4B:** *Friedhoff 2* **Energy & Society**	**Session 4C:** *Friedhoff 3* **Food & Nutrition**
10:15 - 10:30 AM	**Networking Break 7**		
10:30 AM - 12 PM	**Session 5A:** *Friedhoff 1* **Smart Energy Grids**	**Session 5B:** *Friedhoff 2* **Healthcare 2**	**Session 5C:** *Friedhoff 3* **Disaster Prevention & Response 2**
12:00 - 12:30 PM	**Lunch Buffet** *Friedhoff 1*		
12:30 - 1:15 PM	**Panel Session: Educating Engineers for Global Impact - Humanitarian Technologies in the Classroom and Beyond** *Friedhoff 1* Sponsored by Berkeley Development Engineering Program – Gold Patron Facilitated by Yael Perez, University of California Berkeley		
1:15 - 1:45 PM	**Networking Break 8**		
1:45 - 3:15 PM	**Session 6A:** *Friedhoff 1* **Humanitarian Engineering Education**	**Session 6B:** *Friedhoff 2* **Healthcare 3**	**Session 6C:** *Friedhoff 3* **Innovative Technologies**
3:15 - 3:30 PM	**Networking Break 9**		
3:30 - 4:30 PM	**Closing Ceremony & Awards** *Friedhoff 1*		
4:30 - 6:00 PM	**Networking Break 10**		
6:00 - 8:00 PM	**Networking Social Event at the Labriola Innovation Hub** Sponsored by IEEE Young Professionals – Registration Required		
SATURDAY, OCTOBER 25th			
10 AM - 12 PM	**Caffeine & Colleagues** – Registration Required *Labriola Innovation Hub*		
1:45 - 4 PM	**Local Event: Coors Brewery Tour** – Registration Required		

Detailed Program

Wednesday, October 22nd

2:30 PM	**Registration Opens**
3:15 – 4:45 PM	**Workshops**
Friedhoff 1	**Workshop 1: Centering Context and Communities in IEEE Tech4Good** Juan Lucena, *Colorado School of Mines* & Diana Duarte, *Diversa, Colombia*
Friedhoff 2	**Workshop 2: Becoming an Intentional Engineer** Jeff Perry, *Coach/Trainer/Author at jeff-perry.com*
5:00 – 7:00 PM	**Welcome Gathering & Poster Session**
Poster Session Green Center Main Lobby	**A Multi-Agent System for Automatic Charity Recommendation, Financial Audit, Political Activity Check, and Compliance Risk Assessment,** Yunjia Guo, *YK Pao School* **Advancing Rural Water Access Through Prefabrication of AguaClara Treatment Systems,** Margaret Gottfried, *The Ohio State University* **AI for Safety: Developing an Instance Segmentation Model with Mask R-CNN to Identify Hazardous Chemical Spills in Hospitals and Public Areas,** Sirisha Agarwal, Finn Freas, Patrick C. Riley, Samir V. Deshpande, *U.S. Army Combat Capabilities Development Command Chemical Biological Center* **AI-Driven Indoor Garden System,** Ian Rodrigues Kmiliauskis, Yong Lin, *Miami University*, Wei Linjing, *Gansu Agricultural University* **apeSense: Automated Grape-Leaf Disease Detection Using a Deep Learning Pipeline on Augmented Vineyard Imagery,** Sanskriti Farswal, *University of Maryland*, Rishit Agarwal, Purdue University, Navid Shaghaghi, *Santa Clara University* **Assessing Methane Emissions and Energy Recovery Potential in Underground Coal Mining,** Kevin Sumpon, Sergio A. Escudero, Jessie M. Zarazaga, Kathleen M. Smits, *Southern Methodist University* **Automated UV Water Sterilization System for a Guatemalan Elementary School,** Kiera Crawford, Luke Bell, *Colorado School of Mines* **Beyond the Trip: Evaluating Impact Through Humanitarian Study Abroad Projects,** Ogochukwu Elisha-Wigwe, Anastasia Schauer, *The University of Texas at Austin* **Breathe Easy On-Board & Beyond: Real-Time, Energy-Aware Solar IoT Air Monitors—Bus Stop to Cabin,** Mercy Nthenya Kyatha, *University of Massachusetts Amherst* **Bridging or Widening the Gap? Evaluating LLM Performance on Translated vs. Non-Translated Queries in Low-Resource West African Languages,** Ndiana Obot, *Hinsdale South High School* **Campus Sustainability Planning with SPeAR®: Santiago Christian School,** Sawyer Shepherd, *Lipscomb University* **Contextualizing Computational Models: A Case Study on Institutional Trust in the University of Illinois,** Naina Balepur, Hari Sundaram, *University of Illinois Urbana-Champaign* **Contributing factors to successful technology transfer of autonomous systems in agriculture, manufacturing, and medical devices,** Debbie Hodson, Grace Burleson, *University of Colorado Boulder* **Delivering Water Safety for All: Affordable & Continuous IoT-Based Chlorine Monitoring with Lab and Field Evidence from Western Kenya,** Pratiyush Singh, Louis Zannou, Arnita Rut Murnandityas, Ibrahim Ukrin Putra Adi Dharmesta, Victoria Hollingshead, Jeremy Lowe, Amy Pickering, *University of California Berkeley* **Design of a Digital Twin Framework for Predicting Environmental Effects on Frass Fertilizer Quality,** Victor Kimaru, Collins Bett, *International Centre of Insect Physiology and Ecology* **Designing for Resilience: Engineering an Accessible Wash-and-Pack Station to Support Urban Agriculture and Food Equity,** Kasen Holt, *Lipscomb University* **Developing Undergraduate Student Leaders in the Humanitarian Engineering Sector in a Hybrid Course and Organization Model,** Olivia Spencer, Sarah Moore, Paul Kupfer, *Clemson University* **Educating for co-design and place-making: Engineers envisioning and enacting sustainable technologies in Colombia and Tanzania,** Jessie Marshall Zarazaga, *Southern Methodist University*

Embedding Human Rights in Engineering Education: A Global South Case Study on Inclusive Transportation, Janine Siqueira Borges, Davis Chacon-Hurtado, Shareen Hertel, Arash Esmaili Zaghi, Danielle Nadeau, Cory Runstedler, *University of Connecticut*

Engineering Through Political Complexity: Lessons From El Socavón, Elaine Wiles, Natalie Gillum, *Colorado School of Mines*

Exploring Engineers' Strategies and Skills used to Engage External Stakeholders, Emma Balevic, Grace Burleson, *University of Colorado Boulder*

Exploring the Scalability of AguaClara Technology to address Water, Sanitation, and Hygiene (WaSH) Challenges in Rural East Africa: A Comparative Study of Social, Economic and Environmental Contexts, Yvette Onyango, *The Ohio State University*

From Campus to Community: Evaluating Student-Led Humanitarian Engineering Projects Through the SDG Lens, Laura Doyle, Sara Lampshire, Eva Topete, *Santa Clara University*

How are variables of energy system decision making models identified and defined?, Sophie Frey, Grace Burleson, *University of Colorado Boulder*

How context changes technical decision making utilizing a Honduran water infrastructure case study, Emily Lawson-Bulten, *University of Illinois at Urbana-Champaign*

How Universities Can Repurpose E-Waste and Support Underserved Communities, Dylan Fall, Sean Alexander, Luke Mackenski, Yong Lin, *Miami University*

Hydration Automation: A Frugal & Smart IoT Irrigation System, Siddhi Kabadi, Arnav Bharadwaj, Hayoung Ryow, *Santa Clara University*

Hydraulic Modeling for Guatemalan Water System, Sara Berry-Brown, Nolen Ritzel, Lauren Baker, Joelle Noble, Trey Owen, Jessamine Reckard, Kevin Colvett, *Colorado School of Mines*

Improving EMS in Underserved Areas with Drones, Theodore Tikhomirov, *Latin School of Chicago*, Kelly Griendling, *Georgia Tech*

It's Not What You Know, It's Who You Know: Insights on Community Engagement in Rural America, Tristan Clark, Mikhaila Powers, Lauren Ellens, Elizabeth Reddy, *Colorado School of Mines*

Machine-Learning Forecasts of Daily Solar Irradiance for Tribal Energy Planning: A Case Study with the Prairie Band Potawatomi Nation, Hongyu Wu, Maddie Harp, Samuel Largo, *Kansas State University*

Opportunities and Challenges in Low-cost Sensor Development to Support Engineering Design in Data-poor Contexts: Case Studies from Uganda, David Wituszynski, Bruce Arensen, Daniel Propst, *Engineering Ministries International*

Optimizing Residential Energy Strategies by Income Through EnergyPlus Simulations, Hankyu Koo, *Bellarmine College Preparatory*, Hohyun Lee, *San Jose State University*

Power from Earth: Geothermal Rock-Based Energy Solutions for Off-Grid Communities, Clare Otwori, *Dedan Kimathi University of Technology*

Preliminary Battery Charge Capacity Testing Under Varying Load Profiles, Daniel Kasakula, Henry Louie, *Seattle University*

The Design Process and Re-Designing Design Education, Alexandra Fowler, Percy Smith, *University of Colorado Boulder*

The intersection of climate variability, water and food insecurity, and household resilience in Rwanda, Arnold Bugingo, Laura MacDonald, Evan Thomas, *University of Colorado Boulder*, Lambert Mugabo, *Virridy*

Thursday, October 23rd

7:30 – 8:00 AM	**Breakfast**
8:00 – 8:45 AM	**Conference Welcome & Opening Remarks**
8:45 – 9:00 AM	**Networking Break 1**
9:00 – 10:15 AM	**Technical Session 1**
1A: People-Focused Practice Chair: Patrick Sours Room: Friedhoff 1	***Volunteering in Humanitarian Engineering Projects: Motivations and Challenges Experienced by Students, Professionals, and Retirees,*** Emma G Balevic and Grace Burleson ***Embedding the 3Es into Engineering Curricula through the VIP4SD Model at the University of Strathclyde,*** Scott M Strachan, Damien Frame and Thomas Devaney ***Simulating Context: Pathways to User Representation in Global Development Engineering Projects,*** Ogochukwu K.U. Elisha-Wigwe and Anastasia M Schauer ***Engineering in Context: Co-designing Construction, Demolition, and Textile Waste Recycling with Medellín's Recyclers,*** Mariana Vélez-Fernández, Noah M Link, Shelby Mary Gilson, Danielle Kosowsky and Juan Lucena ***Co-Constructing Sustainable Coastal Livelihoods in Tamil Nadu: A Community-Centred Conservation Framework,*** Reshma Ramesh, Sruthi M, Lekshmi S and Mithravinda S P
1B: Agriculture & Water Chair: Min Pack Room: Friedhoff 2	***Diagnosing Community Readiness for Effective Water Governance Using Machine Learning: A Pilot from Rural India,*** Mandida Gusha, Sreelakshmi R, Suryaparvathy S, Victoria Ustenko, Srividya Sheshadri, Reshma Ramesh and Bhavani R Rao ***AI-Enabled Sapling Health Monitoring for Climate-Resilient Agriculture: A Pilot Deployment from Rural Odisha,*** Balu M Menon, Ayyappan Ajan, Ramakrishnan K, Gokul Dev B, Pranav Shaji, Lekshmi Venkitesh and Rao R. Bhavani ***Leaking Potential: Uncovering Gaps in Water Technologies and Techniques Adoption in Colorado's Small-scale Farming Through Asset-Based Analysis,*** Alexandria Rodgers and Marie Stettler Kleine ***On Recommending Sustainable Agricultural Practices in Rice Cultivation with a Digital Twin Model,*** Rushikesh Dattatraya Kulat, Vidit Patil, Sanat Sarangi, Dineshkumar Singh and Srinivasu Pappula ***Challenges at Each Stage of the Kenyan Cage Fish Farming Cycle,*** Lilian Downs, Anshu Nunemunthala, Gauri Patil, Adeline Weitknecht, Benjamin Roy and John Gershenson
1C: Disease Prevention & Care Chair: Khanjan Mehta Room: Friedhoff 3	***A data-driven mode for public health interventions: a case study of the 2025 measles outbreak as reflected in social media,*** Han Kyul Kim, Andy Skumanich, Reza Davariar, Ryaan Sayegh, Christopher J Royse, Shelly Wei, Lavanya Sharma and Vishnu Sitaram Rajeev ***Stigma to Screening: Generative AI and Self-Sampling Tools to Expand Cervical Cancer Screening in Low- and Middle-Income Countries,*** Lorraine Rwasoka, Jackson Kramp and Khanjan Mehta ***Strengthening the Sickle Cell Care Pipeline: A Systems-Based Approach in Sierra Leone,*** Owen R Holst, Nancy D Cárcamo-Paiz, Adelaide K Ogden, Nicole Hirsch, Amelie Fry, Jacob M Feuerstein, Xuanhong Cheng and Khanjan Mehta ***TrackTB: An Accessible Solution for Improved Tuberculosis Treatment Adherence in the Philippines,*** Maia R Wainwright, Sanvi Bhatnagar, Kyle T Smith, Francisco Villa, Barbara Monteiro-Carvalho, Sean Vassilaros and Khanjan Mehta
10:15 -10:30 AM	**Networking Break 2**
10:30 AM – 12 PM	**Technical Session 2**
2A: Disaster Prevention & Response 1 Chair: Elizabeth Reddy Room: Friedhoff 1	***DTN7-ESP: A native DTN-based Communication System for ESP32 Microcontroller,*** Julian Zobel, Tim Öhlenschläger and Björn Scheuermann ***On the field-readiness of LoRaWAN for Weather Station networks in Developing Regions,*** Rytis Paškauskas and Marco Zennaro ***FloodGuardian: A LoRa-Based Early Flood Warning System for River Basins,*** Santiago Pereyra, Cristina Mayr ***Interpretable Dual-Stream Learning for Local Wind Hazard Prediction in Vulnerable Communities,*** Mahmuda Akhter Nishu, Chenyu Huang, Milad Roohi and Xin Zhong ***AI-Driven Data Fusion for Humanitarian Visual Analytics: Integrating Multimodal Social Media Insights on Ukrainian Refugee Displacement,*** Brian Tomaszewski, Natalie Crowell and Olivia Croteau

	AI monitoring of social media to assist in crisis preparedness and response: COVID-19 Case Study, Andy Skumanich, Han Kyul Kim, Nick Sawyer, Reza Davariar, Jahan Dokht Hill and Michael Kan
2B: Community-Centered Education Chair: Juan Lucena Room: Friedhoff 2	***A Culturally Adapted Modified Guided English Reading Program for K-1 Students in India,*** Prisha M Juddoo, Neeraja S, Akshaya S, Punya P, Georg Gutjahr, Radhika Menon, Prema Nedungadi and Raghu Raman ***Analysis of Secondary Schools' Lesson Planning Processes in Kisumu, Kenya,*** Christina Sang, Jaiden Asch and John Gershenson ***A Three-Dimensional approach to Education and Development of Rural India,*** Ranjan K Sen, Bulbul Sen and Indira Dasgupta Cherukuri ***Development of a Digital Literacy Curriculum for Rural Women in India,*** Drishti Marwaha, V R Bhanu, Sayujya R S and Sruthy V Nair ***Development of a Life Enrichment Education Curriculum for Rural Women in India,*** Drishti Marwaha, V R Bhanu, Nandha S and Alden Philip ***Scaling Humanitarian Design and Social Enterprise (HDSE) Education in India: The HDSE Virtual Academy and Leadership Intensive,*** Megan Thomas, Juliana Magarelli, Callie Higgins, Gabriela Quinteros, Eric R Obeysekare and Khanjan Mehta
2C: Context-Driven Approaches (*Lightning Talks*) Chair: Julia Kramer Room: Friedhoff 3	***Agentic Educational Content Generation for African Languages on Edge Devices,*** Ravi Gupta and Guneet S Bhatia ***Using Open Access Remote Sensing to Improve Grazing Access and Reduce Conflict Among Pastoralists in Northern Kenya,*** Matthew Leopold, Audrey Xiao, Vincent Park, Catherine McManus, Ahmed Mohamed, Siddhartan Govindasamy and Justin Henriques ***Participatory Methods for Relational Power Mitigation and Increased Contextual Understanding in Rural, Infrastructure Design,*** Emily Lawson-Bulten, Connor LaSalle and Ann-Perry Witmer ***Transforming Tsunami Footage into Experiential Learning for Disaster Innovation,*** Emanuel Leleito ***Interdisciplinary Co-Design & Research for Humanitarian Design: Promise of Inclusion or Exclusion?,*** Rachel Howell ***Beyond Coding: Human-Centered Design in Computer Engineering Senior Projects: Lessons from the NicaAgua App,*** Allan Baez Morales, Iris Stewart-Frey, Qiuwen Li, Angela Musurlian and Nicolas Gibson ***Understanding Public Perceptions and Adoption of Community Parametric Reciprocal Disaster Insurance in Sarasota County, FL,*** Margaret L Miles, Julia Kramer, Guillermo Franco and Sabine Loos ***Developing and Leveraging Project Management Frameworks to Maximize Impact in Undergraduate Humanitarian Engineering Practice,*** Paul A Kupfer

12:00 – 12:30 PM	**Lunch Buffet**
12:30 – 1:15 PM	**Panel: When Hardware Meets Software: How Humanitarian NGOs Balance Social Dimensions with Technical Interventions** ***Sponsored by Mines Humanitarian Engineering Program – Platinum Patron*** • Sam Temple, *Colorado School of Mines* • Max Churchfield, *iDE* • Andrew Foote, *Sanivation* • Bruce Pelz, *Maji Safi Group* • Melissa Montgomery, *Engineers Without Borders USA*
1:15 – 1:45 PM	**Networking Break 3**
1:45 – 3:15 PM	**Technical Session 3**
3A: Participatory Design Chair: Anastasia Schauer Room: Friedhoff 1	***Assessing Partnership: Utilizing Community-Based Participatory Research to Evaluate Participatory Design,*** Lauren R Wojciechowski and Julia Kramer ***Impacts of AguaClara OSU- Collaboration in the RIDE (Research, Innovate, Design, and Empower) Ecosystem Focused on Passive Gravity Water Treatment,*** Patrick J Sours and Maggie Gottfried ***Design for Humanitarian Markets Revisited: Human versus Artificial Accountability,*** Brita Fladvad Nielsen and Ana Laura Santos ***The Design and Testing of a Low-cost Prosthetic Hand for use in Jaipur, India,*** Steven Reimer, Christopher Kitts and Michael Neumann

	Pumps and Pipes are Easy; People and Politics are Hard - A Case Study of a Rural Water System in Guatemala, Kevin Colvett ***Cultivating Future Innovators: A Review of India's Atal Tinkering Labs,*** Emma Najarian, Lilia Gana, Arpita Zaman, Theodore Spear, Nadxieli Jimenez Bielma, Khanjan Mehta and Eric R Obeysekare
3B: Healthcare 1 Chair: Junko Munakata Marr Room: Friedhoff 2	***From Lab to Field: Lessons from a Design-Led PhD in Humanitarian Healthcare Technology,*** Md Haseen Akhtar ***Design of an Edge-based Portable EHR System for Anemia Screening in Remote Health Applications,*** Sebastián A. Cruz Romero, Misael J. Mercado Hernández, Samir Y. Ali Rivera, Jorge A. Santiago Fernández and Wilfredo E. Lugo Beauchamp ***Challenges and Dynamics of Electronic Health Record (EHR) Appropriation in Senegal: A Literature Review and Field-Based Investigation,*** Papa Alioune Cissé, Marie Ndiaye, Omar Sow, Papa Ibrahima Ndiaye, Abdoulaye Ngom and Bala Biaye ***Barriers and Facilitators to the Implementation of an Evidence-Based Digital Doula Postpartum Peer Support into a Sustainable Medicaid Reimbursement System in Rural New England,*** Sai S Chilakapati, Jillian L. Berch, Shahed Khudaier, Margaret R. Emerson, Alka Dev, Daisy J. Goodman and Karen Fortuna ***The Value of Fine Tuning a Maternal Health AI Chatbot with Local Country Expertise,*** Aidan Chen, Christopher Zhang, Aiza Dahman, Lara Barroco, Annalicia Pickering and Milton Chen ***NLP-Driven Emotion Recognition for Individuals with Impaired Speech,*** Riya Jain, Fiona Keaney and Kanika Sood
3C: Economic Empowerment Chair: John Gershenson Room: Friedhoff 3	***Feasibility of Improving the Financial Security of Boda-Boda Drivers Through a Sustainable Business Model--Part I,*** Rosie Seidenberg, Sophie Wright, Purva Gupta, Lauren Wagner and John Gershenson ***Feasibility of Improving the Financial Security of Boda-Boda Drivers Through a Sustainable Business Model--Part II,*** Rosie Seidenberg, Sophie Wright, Purva Gupta, Lauren Wagner and John Gershenson ***Leveraging GIS for Strengthening Self-Help Groups (SHGs): A Tool for Inclusive and Sustainable Development,*** Liji B, Karthik Sankaranarayanan and Sajithkumar K J ***LokalFi: A Framework for Long-Range Wi-Fi Deployment in the Philippines,*** Riza Carmela M Pineda, Philip A. Martinez and Kenneth Rey Sumalinog ***The Impact of Using AI in Research and Development in Non-Profit Organizations in Saudi Arabia,*** Bandar F Alzahrani ***Towards Safe Commute: An AI Driven Fusion Model For Real Time Forecasting and Emergency Alerting,*** Krishna Mohan Bathula, Sai Teja Bainapalepu and Aditya Bhavsar
3:15 – 3:30 PM	**Networking Break 4**
3:30 – 5:00 PM	**Featured Sessions**
Friedhoff 1	***Lecture:*** **Current US Policy Trends Shaping Technology,** Erik Heilman, IEEE USA
Friedhoff 2	***Workshop*****: Grid Down Scenario Table Top Exercise,** David Vaughn, Clemson University
Meet in Green Center Lobby	***Tour*****: Colorado School of Mines Labriola Innovation Tour**
5:00 – 5:30 PM	**Networking Break 5**
5:30 – 7:30 PM	**Dinner Banquet** ***Keynote:*** **Dr. Ann-Perry Witmer, University of Illinois: "A Self-ish Approach to Technology in Context"**

Friday, October 24th

7:30 – 8:15 AM	**Breakfast**
8:15 – 9:00 AM	***Featured Session*: Educational Resources to Deploy Technology for Social Impact and Their Gaps: An Interactive Workshop** ***Sponsored by IEEE HTB – Platinum Patron*** Led by Stephanie Gillespie, IEEE Educational Activities Board
9:00 – 9:15 AM	**Networking Break 6**
9:15 – 10:15 AM	**Technical Session 4**
4A: Data-Driven Decisions Chair: Ann Perry-Witmer Room: Friedhoff 1	***Digital Monitoring for Health Facility Emission Prediction and Climate Finance: A Case Study from the Democratic Republic of the Congo,*** *Daniel Steven Doxazo Sitompul, Samuel Benjamin Miles, Rashmi Varma, Sylvia Imanirakiza, Sanuratu Koroma, Jackson Mughuma, Ryan McCord, Olufolahan Osunmuyiwa, Janvier Kamundala, Ngalula Sandrine Mubenga, Jay Taneja, Daniel Merson Kammen and Laura H Kwong* ***Global Distributed Tracking of Supplies via Free-libre Open Source Cryptography Obviating Authentication,*** *Robert L Read, Victoria Jaqua, Christina A. Cole, Francesca Williams, Nina Lahoti and Nora Moor* ***Quantifiable Theory of Change,*** *Franz Busse, Chris Mattson and Truman Spring* ***Applications of In Situ Microbiological Water Quality Monitoring,*** *Whitney N Knopp, Joshua T Klaus, Laura MacDonald and Evan Thomas*
4B: Energy & Society Chair: Henry Louie Room: Friedhoff 2	***A Human-Centered Incentive Design for Equitable Grid Resilience in Disadvantaged Communities,*** *Chelsea Okwechime and Hamid Nazaripouya* ***Sociotechnical Development of Offshore Wind: A Case of Kitty Hawk North,*** *Mikayla L Friday and Desen S Özkan* ***Review of Technical and Non-Technical Key Performance Indicators for Microgrids Used for Rural Energy Services,*** *Rabia Khan, Ian Baring-Gould, Noel Schulz and Liadi A Akande* ***Relationships as Engineering Infrastructure: Scaling Community-Based Food, Energy and Water Solutions Through Human Networks,*** *Yael Valerie Perez, Cheryl Schwab, Kathy Isaacson, Peter Romine, Dustina Gill, Donna Brown, Karletta Chief and Alice Agogino*
4C: Food & Nutrition Chair: Khanjan Mehta Room: Friedhoff 3	***Towards Low-Sodium Diets: Industrial Approaches to Advancing Public Health in Underserved Communities,*** *Daniel B Weir, Elizabeth Ehrhardt, Joseph A Menicucci, Jr and Khanjan Mehta* ***Bouillon Cubes as Vehicles for Micronutrient Delivery: A Systematic Review of Fortification Approaches and Productization Opportunities,*** *Samantha R Sandhaus, Shelly Deng and Khanjan Mehta* ***Front-of-Package Labelling as a Global Public Health Tool: A Comprehensive Review of Nutritional Labelling Systems and Their Role in Reducing Diet-Related Non-Communicable Diseases,*** *Elizabeth Ehrhardt, Laurent Tong, Tara Chelvam, Khanjan Mehta and Joseph A Menicucci, Jr* ***Evaluation of Productivity and Environmental Impact of Hybrid Sweet Corn Varieties with Digital Twins,*** *Ruturaj Nivas Patil, Abhishek Kumar, Sanket Junagade, Rushikesh Dattatraya Kulat, Swagatam Bose Choudhury, Sanat Sarangi, Suryakant Sawant, Nandan Rajpoot, Ajay Mittal, Dineshkumar Singh and Srinivasu Pappula*
10:15 - 10:30 AM	**Networking Break 7**
10:30 AM – 12 PM	**Technical Session 5**
5A: Smart Energy Grids Chair: Dean Nieusma Room: Friedhoff 1	***Optimal Resource Management in Peer-Connected DC Microgrids for Off-Grid Communities,*** *Rabia Khan and Noel Schulz* ***Smart Local Energy Systems for Grid Densification and the Alleviation of Load Shedding: Malawi Case Study,*** *Damien Frame, Scott M Strachan and Lewis Carty* ***Effects of Load Estimation Detail on Off-Grid System Design and Reliability,*** *Melvin Dang, Dean Aquino and Henry Louie* ***Case Study of the Cheeba Solar Energy Kiosk,*** *Peter Dauenhauer, Rabia Khan, Bennett Schmitt, Hansel Mweetwa, Brad Helland and Henry Louie* ***Accelerated Network Design for Analyzing Spatial Heterogeneity in Electrification Planning,*** *Yuezi Wu and Vijay Modi*

	IV Curve Fitting Using Machine Learning Techniques and Low-Cost Technologies for Data Acquisition, *Jimmy Cordova, Manuel Alvarez-Alvarado, Ivan Endara, Fernando Vaca-Urbano, Augusto Carpio Astudillo and Kevin Garcia Gonzales*
5B: Healthcare 2 Chair: Cindy Athanasiou Room: Friedhoff 2	***Bridging Neurological Care Across Borders: A Comparative Study of Kazakhstan and U.S. Healthcare Systems,*** *Angelina Penza, Tiya Ketul Patel, Jonathan P Sferratore, Emma J Jovanovic, Luke H Blum, Muza Yelgezek, Daiyana Jumagalieva and Dhruv R. Seshadri* ***Designing an Off-Network Paging System to Improve Continuity of Patient Care in Low-Resource Settings,*** *Olivia M Meyer, Regina Ruiz, Lillian Wu, Meghan G Young, Annly John, Kavya Famolari, Victoria Kuzma, Marianna Angle, Eric R Obeysekare and Khanjan Mehta* ***Pack . Deploy . Heal: Reimagining Healthcare Delivery for Resource Constrained Settings Through Design Innovation,*** *Md Haseen Akhtar, Kirk Mendoza, Mark Anderson, Peter Anderson and J Ramkumar* ***Translational Telemedicine for Indigenous Health: A Community-Centered Approach in the Karmata Rúa Reservation,*** *Hernan Felipe Garcia Arias, Laura Garcia-Mosquera, Luis Fernando Torres-torres, Alexandr Passo-Polanco, Isaac Uribe-Jaramillo, Luisa Taho, Weimar Arenas Gonzalez, Maria Clara Marín-Valencia, Alejandro Hernandez-Arango, Sara Valencia-Cadavid, Augusto Salazar and Jesus Vargas Bonilla* ***Assessing Stroke and Dysphagia Care Practices in Kazakhstan,*** *Nicholas Evaristo, Muza Yelgezek, Daiyana Jumagalieva and Dhruv R. Seshadri* ***Cost-Effectiveness of a Menstrual-Hygiene Health-Education Program for Tribal Adolescent Girls in Rural India,*** *Radhika Menon, Merin Mathew, Georg Gutjahr, Prema Nedungadi and Raghu Raman*
5C: Disaster Prevention & Response 2 Chair: Alina Handorean Room: Friedhoff 3	***Coordinated Autonomous Drones for Human-Centered Fire Evacuation in Partially Observable Urban Environments,*** *Maria Mendoza, Addison Kalanther, Daniel Bostwick, Emma Stephan, Chinmay Maheshwari and Shankar Sastry* ***Hazardous materials storage, transportation, and disposal: A critical analysis of green supply chain methodologies,*** *Santiago Neira Mendieta, Diana Mercedes Rodriguez Coca, Tony McAleavy and Lisa Wier* ***Humanitarian demining: comparing diurnal landmine thermal detection simulations with real field tests,*** *Alberto Martinetti, Ben Oregan, Amir Shah, Reinis Jaunarajs, Sven Askes and Francesco Nex* ***Two-Decade Assessment of Land Degradation (SDG 15.3.1) and Urban Expansion in Lucknow Using MODIS and Landsat,*** *Zahra A Rizvi* ***Using Retrieval Augmented Generation and Knowledge Graphs to Understand Climate Obstruction,*** *Michael DeBellis, George Gino, Aadarsh Balaji and Jacob Gino* ***A Humanitarian Perspective on the Uncrewed Aerial System Design Process,*** *Carina Koutsambasis, Nick J Brown and Matthew Marino*
12:00 – 12:30 PM	**Lunch Buffet**
12:30 – 1:15 PM	**Panel: Educating Engineers for Social Impact: Humanitarian Technologies in the Classroom and Beyond** ***Sponsored by Berkeley Development Engineering Program– Gold Patron*** • Yael Perez, *University of California Berkeley* • Alice Agogino, *University of California Berkeley* • Donna Brown, *American Indian Higher Education Consortium* • Kirsten Dodson, *Lipscomb University* • Laura MacDonald, *University of Colorado Boulder*
1:15 – 1:45 PM	**Networking Break 8**
1:45 – 3:15 PM	**Technical Session 6**
6A: Humanitarian Engineering Education Chair: Laura MacDonald Room: Friedhoff 1	***A Model for Utilizing a Global Capstone Course to Strengthen International Humanitarian Engineering Programs,*** *Adithya Jayakumar and Patrick J Sours* ***Co-Defining Success and Impact of Co-Creating through International Engineering Projects,*** *Philippe Damoiseaux, Taylor Ericson, Colleen McCulloch, Melissa Trykowski and Marie Stettler Kleine* ***Opportunities and Challenges in Leading a Community-Engaged International Research Experience for Students (IRES) Program,*** *Stephanie A. Claussen, Kirsten A Davis and Jennifer DeBoer* ***Exploring Global Responsibility During an Immersive International Design Experience,*** *Olivia Wilburn, Alexandra Gartrell and Grace Burleson*

	Humlog Action Lab: practice in higher education, *Diana Mercedes Rodriguez Coca, William Guerrero, Elyn Solano-Charris and Holman Ospina-Mateus* ***From Classroom to Community: Engaging Students and Community Partners in Co-Design,*** *Benjamin M Phillips*
6B: Healthcare 3 Chair: Alina Handorean Room: Friedhoff 2	***An Improved Health Information Management System for Korle-Bu Teaching Hospital in Ghana,*** *Jochebed Afua Basil, Aaron Tsastu Tamakloe and Dennis Asamoah-Owusu* ***Real-Time Surgery, Delayed: Internet Latency and the Prospect of Telerobotic Surgery,*** *Kennedy O Orwa, Yixin Chen, Varun Bathija and Mike H.M. Teodorescu* ***Healthcare Alliance for an Equitable World (HAEW): Bridging Global Gaps in Healthcare,*** *Kim Tranquada* ***Development of a Hearing Probe for a Low Cost Newborn Hearing Screening Device,*** *Venkadesh Eswaranandam, Ayush Chakraborty, Elizabeth Johansen, Bradley A Minch and Siddhartan Govindasamy* ***Iterative Development of the Mothers of Sierra Leone Cinema Kit: A Culturally-Responsive Humanitarian Engineering Approach to Improving Maternal Health,*** *Eileen J Kandie, Chloe Chan, Julia E Killar, Madeline M Boughner, Eesha Gudoor, D. Michael Kramp, Fathima Wakeel and Rejoice Obiora* ***Igloo Mobile Primary Health Center: A Systematic Design-Led Approach,*** *Md Haseen Akhtar and J Ramkumar*
6C: Innovative Technologies (Lightning Talks) Chair: John Gershenson Room: Friedhoff 3	***A Low-Cost Multispectral-Camera System,*** *Angel Rivas and Siddhartan Govindasamy* ***The Infestation of the Red-Billed Quelea Bird and the Infeasibility of a Suitable Deterrent System,*** *Nicholas Ene, Catell Le Baut and John Gershenson* ***Impact of Variable Water Conditions on Cage Fish Farming in Lake Victoria,*** *Lilian Downs, Anshu Nunemunthala, Gauri Patil, Adeline Weitknecht, Benjamin Roy and John Gershenson* ***3D-printed clay-based biosand filters,*** *Ethan Iams, Hannah Salazar, Jason Kooiman, Michaela Horst and Min Pack* ***SmartCane: Intelligent Indoor Navigation for the Visually Impaired,*** *Dongbin Lee, Joao Pedro De Favari Engel, John Milton A Ponce, Geoffrey Gonzales, Brian Ngo and Pramod Gupta* ***Multimodal low-cost solutions for Air Quality Index estimation,*** *Kapilan Karunakaran and Seren Smith* ***Fabrication of an Air-Assisted, "Pelicanized" Lab for Disaster Preparedness and Response,*** *Kourosh Arasteh, Matt Lyman, K. Wesley Overton and Kristl Adams* ***An Automated Percolation Tester for the Determination of Long-term Application Rates of Wastewater to Diverse Soils,*** *David M Wituszynski, Bruce Arensen, Karl Edler, Nathan Wiebe, Jessica Carlson, Cody Esau, Daniel Propst and Lawrence Moos* ***Advancing Campus Sustainability through Waste Sorting and Value Addition at Gulu University,*** *Paige Balcom*
3:15 – 3:30 PM	**Networking Break 9**
3:30 – 4:30 PM	**Closing Ceremony & Awards**
4:30 – 6:00 PM	**Networking Break 10**
6:00 – 8:00 PM	**Networking Social Event at the Labriola Innovation Hub** Sponsored by IEEE Young Professionals (Registration Required)

Saturday, October 25th

10:00 AM – 12 PM	**Caffeine & Colleagues** (Registration required)
1:45 – 4:00 PM	***Local Event:*** **Coors Brewery Tour** (Registration required)

Overview of Workshops & Special Sessions

Workshop 1 (Wednesday, 3:15-4:45pm): Centering Context and Communities in IEEE Tech4Good

Speakers: Juan Lucena, Colorado School of Mines & Diana Duarte, Diversa, Colombia





Overview: This interactive session shows how to center context and communities in Engineering-for-Good (E4G) student competitions using IEEE Tech4Good as an example. Through the examples of IEEE Tech4Good-Colombia and -Andean Group, we show how these competitions can 1) engage communities from the beginning in problem definition and solutions; 2) include community-engagement partners as bridge builders between teams and communities; 3) reach out and engage engineering schools and communities in the most vulnerable regions of a country; and 4) provide the processes and institutional support to continue serving a country's most vulnerable communities. One of the most salient challenges that E4G student competitions face is encouraging students to acquire empathy, contextual knowledge, and epistemic humility to co-define problems and solutions with the communities they hope to serve. Often, student teams enter these competitions without fully understanding the context (historical, political, institutional, cultural, etc.), the diversity of stakeholders, the complexities within and among communities, the full dimensions of the opportunities and limitations available, and the unintended consequences of their assumptions, tools, and engagement practices involved. Using the examples of IEEE Tech4Good-Colombia and -Andean Group, this session aims at contributing to the evolution and maturing of E4G competitions, which in many cases are the starting points of the education of socially responsible engineers and of the relationship building between engineering programs and solutions with the communities they hope to serve. Often, student teams enter these competitions without fully understanding the context (historical, political, institutional, cultural, etc.), the diversity of stakeholders, the complexities within and among communities, the full dimensions of the opportunities and limitations available, and the unintended consequences of their assumptions, tools, and engagement practices involved. Using the examples of IEEE Tech4Good-Colombia and -Andean Group, this session aims at contributing to the evolution and maturing of E4G competitions, which in many cases are the starting points of the education of socially responsible engineers and of the relationship building between engineering programs and communities.

Workshop 2 (Wednesday, 3:15-4:45pm): Becoming an Intentional Engineer



Speaker: Jeff Perry, Coach/Trainer/Author at jeff-perry.com

Overview: Who do you want to become? Building a meaningful engineering career is about more than just landing a job and climbing the ladder. It's about being intentional about what you want to learn, the impact you want to have, and how you want to grow. With many potential paths for professionals, charting our path can be confusing and difficult. This workshop can help you get started on making it easier. Objectives: Learn what intentionality is and how it is the key to a career of impact and satisfaction; Discover how our mindsets drive our actions and behaviors, and some ideas of how to shift your own mindset; Develop a new approach to networking that is more about building relationships than collecting contacts or followers.

Panel (Thursday, 12:30-1:15pm): When Hardware Meets Software: How Humanitarian NGOs Balance Social Dimensions with Technical Interventions

Facilitator: Sam Temple, Colorado School of Mines

Panelists:

- **iDE** (Max Churchfield, Senior Design Manager): Offers affordable technologies to rural farmers in Africa, Asia, and Central America
- **Sanivation** (Andrew Foote, CEO & Co-Founder): Provides waste management technologies and systems for secondary cities across East Africa
- **Maji Safi Group** (Bruce Pelz, President & Co-Founder): Expands access to safe drinking water and sanitation in East Africa
- **Engineers Without Borders USA** (Melissa Montgomery, PE, Chief Programs Officer): Supplies engineering support and technology interventions in North America, Africa, Asia, and Central/South America

Overview: Colorado is home to more than 200 humanitarian assistance organizations, ranging from small, context-specific groups to large-scale global missions. These organizations support initiatives aligned with the Sustainable Development Goals, addressing issues from poverty alleviation to climate action. This panel will feature 4 Colorado-based NGOs that specialize in technology-enabled interventions aimed at improving livelihoods in communities around the world. The participating organizations will share context-specific examples of how technology is being used in the field and will discuss the challenges and opportunities that arise when navigating social, cultural, environmental, economic, and political factors. Rather than focusing on "best practices," the panel will offer an honest exploration of both the successes and failures in implementing humanitarian technologies globally.



Special Session (Thursday, 3:30-5:00pm): Current US Policy Trends Shaping Technology



Speaker: Erik Heilman, IEEE USA

Overview: IEEE-USA Government Relations will provide an overview of policy priorities relevant to technologists. This session will discuss opportunities and challenges for those seeking to align technology solutions with evolving policy landscapes.

Special Session (Thursday, 3:30-5:00pm): Colorado School of Mines Labriola Innovation Tour

Overview: Join us for a tour of the brand new Labriola Innovation Hub - the heart of design and creativity of Colorado School of Mines! The space is home to a metal shop, wood shop, composite and paints shop, electronics shop, and a maker space with over 30 3D printers. Join us on your conference break to peak your creativity!

Workshop 3 (Thursday, 3:30-5:00pm): Grid Down Scenario Table Top Exercise



Speaker: David Vaughn, Clemson University

Overview: The Grid Down Scenario workshop provides a unique, interactive platform for exploring humanitarian technology solutions under extreme conditions. It bridges academic research, community engagement, and technological innovation by immersing participants in a real-world-inspired crisis. The session will demonstrate how resilience planning, GIS tools, and collaborative decision-making can empower underserved communities. By fostering experiential learning and ethical considerations, it highlights the urgency of disaster preparedness in our interconnected world. This workshop will engage diverse stakeholders and leave them with actionable strategies for enhancing community resilience and equity.

Keynote (Thursday, 5:30-7:30pm): A Self-ish Approach to Technology in Context



Speaker: Ann-Perry Witmer, University of Illinois

Overview: We're all aware that technology alone won't solve people's problems. The West spends billions of dollars on technology that fails to improve humanity's conditions, and the failure most often occurs because the technology doesn't match the needs, capabilities, values, and limitations of the people for whom it's created. Engineering tries to address this contextual failure by refocusing design on the user, but our research indicates before we refocus the designs, we need to address the mindsets of the designers. We're talking revolution here – a paradigm shift in approach to acknowledge the power, influence, and participation of the designer in solving the world's greatest physical needs.

Featured Session (Friday, 8:15-9:00am): Educational Resources to Deploy Technology for Social Impact and Their Gaps: An Interactive Workshop



Speaker: Stephanie Gillespie, IEEE Educational Activities Board

Overview: This session will highlight IEEE Humanitarian Technologies' focus on education materials and training for humanitarian engineering project practitioners. The session will be designed to be interactive, with questions posed to the audience to gauge interest in IEEE HT's programs, educational resources, and ways to get involved. The audience will also have opportunities to provide feedback on what educational resources for starting or supporting humanitarian engineering projects may not already be readily available that IEEE HT could develop and provide.

Panel Session (Friday, 12:30-1:15pm): Educating Engineers for Global Impact - Humanitarian Technologies in the Classroom and Beyond

Facilitator: Yael Perez, University of California Berkeley

Panelists:

- **Paige Balcom**, Gulu University in Uganda
- **Alice Agogino**, University of California at Berkeley
- **Donna Brown**, American Indian Higher Education Consortium
- **Laura MacDonald**, University of Colorado Boulder
- **Kirsten Dodson**, Lipscomb University

Overview: This panel will explore how engineering education can better prepare the next generation of engineers to design, develop, and deploy appropriate technologies in low-resource and underserved communities. Drawing from diverse academic programs, panelists will examine strategies for integrating human-centered design, community engagement, interdisciplinary collaboration, and sustainability into engineering curricula. Through this panel, we aim to collectively reflect on this trend—examining its potential, challenges, and impact—and consider how we might shape the future of humanitarian engineering education as a recognized and robust area within both academic and professional practice.

